GENCO meets the neutral wire: four-wire dynamic operating envelopes for low-voltage grids

Extending GENCO to three-phase four-wire low-voltage distribution networks: what breaks, and how a phase-aware extension repairs it

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In brief. GENCO models the grid as balanced, but European low-voltage networks are three-phase four-wire: every home sits on one phase and the neutral carries the imbalance. On a 200-customer benchmark feeder, envelopes from a balanced model met the directional four-wire test in none of the evaluated hours. We found three bottlenecks of GENCO at LV, built a phase-aware GENCO-style model with an explicit phase-and-neutral representation, and used it as a pre-screen in front of a full four-wire OpenDSS check: six-fold fewer OpenDSS simulations, with zero observed false-safe decisions.

The problem

A **dynamic operating envelope** (DOE) gives each household a time-varying import and export allowance instead of a fixed connection limit, so that EVs, heat pumps and rooftop PV fit on the existing network.

- **Network:** open-source benchmark ENWL1 (three-phase four-wire, UK), 800 kVA, 200 single-phase customers, neutral earthed only at the transformer (TT).
- **Households:** Fluvius open data, 2024 quarter-hourly meters; 30% heat pumps, 30% EVs, 40% PV.
- **Limits:** −6%/+10%, i.e. 216.2–253.0 V (UK ESQCR).
- **Test:** an envelope passes if ≥196 of 200 design scenarios meet the directional OpenDSS limits (export: overvoltage, loading; import: undervoltage, loading; full uptake).

Three bottlenecks of GENCO at LV

The stock model did worse than predicting each bus's average training voltage. We found and fixed:

- **Impedance magnitude:** the edge encoder's layer normalisation suppresses the admittances (10× admittances change the embedding by 0.2%). Fix: asinh-scaled features.
- **Physics-loss collapse** at LV scales. Fix: a residual head on a linear power-flow estimate, initialised at zero.
- **Ill-conditioned flows** from voltage differences. Fix: Kirchhoff-current-law flows.

Result: voltage error **1.0e-3 pu**, 5.4× below the physics estimate. We also found and fixed two bugs in the GridFM tooling (best-model export; unpinned slack in datakit).

Operational value

4.81 MWh
export envelope-hours / household / yr

2.50 MWh
import envelope-hours / household / yr

- 36 evaluation days of 2024; day-bootstrap intervals 4.35–5.30 (export) and 2.11–2.87 MWh (import). A fixed allowance valid in every hour would be zero.
- Balanced-model envelopes are 2–3× larger and fail the four-wire check.

Phase-aware GENCO-style model as a pre-screen

A GENCO-style four-wire residual GNN (GENCO's processor family, explicit phase and neutral, 4×4 impedances, no Kron reduction) corrects a one-step four-wire sweep. On 2,968 evaluation states, under the same final check:

35.8%→5.8%
states sent to OpenDSS

0
observed false-safe decisions

Voltage error falls by 91%. Balanced screens, GENCO included, send 63% of states to OpenDSS, supporting the importance of phase information.

Transfer of GENCO representations

- **Balanced embeddings** in the four-wire model: fallbacks 5.8%→11.8%. They do not carry four-wire information.
- **Official pretrained GENCO** on case118: fine-tuning beat training from scratch by **+11.4 points** of reference hosting, in 5 of 5 seeds, at 100 labels.

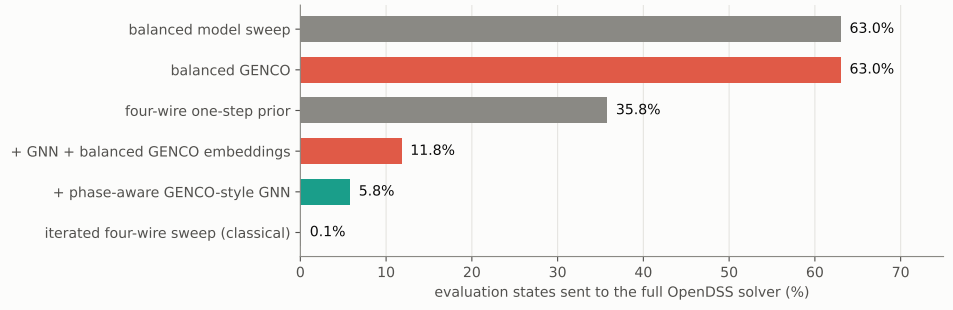
Classical solvers and limitations

- An iterated four-wire sweep remains the accuracy leader (about 80× more accurate). The learned screen's value is six-fold fewer OpenDSS simulations.
- A directional design-set test, not a certificate. Not checked: opposite-side voltage (seen in 12.7% of export and 24.8% of import intervals in at least one fresh scenario), neutral current, partial uptake, unbalance. One feeder.

Open science

- Public inputs (ENWL1, Fluvius open data); pre-registered core evaluation, frozen protocols, disclosed amendments and incidents.
- 84 tests; `reproduce_check.py` regenerates the aggregate reports from retained run artifacts (not yet redistributed); one-minute public demo.

Outlook: LV envelopes sum to the flexibility a DSO offers upstream; one model across LV, MV and HV needs the three changes identified here.



Share of 2,968 evaluation states each screen sends to OpenDSS, all under the same final check; none made a false-safe decision.